WORKED SOLUTION

STEP I 2012, Question 8

Pure • Homogeneous Differential Equations

STEP I 2012, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q8

The key idea

Both equations are *homogeneous*: every term has the same total degree in x and y, apart from an overall power of x. If you put $y = xv$, that power of x cancels and the equation separates into a v-part and an x-part. Part (i) walks you through this. Part (ii) leaves it to you: dividing through by x shows it's the same kind of equation, and the v-integral then needs partial fractions.

Solution

Part (i)

If $y = xv$, the product rule gives $\dfrac{\mathrm{d}y}{\mathrm{d}x} = v + x\dfrac{\mathrm{d}v}{\mathrm{d}x}$. Substituting,

$$x(xv)\left(v + x\frac{\mathrm{d}v}{\mathrm{d}x}\right) + x^2v^2 - 2x^2 = x^3v\frac{\mathrm{d}v}{\mathrm{d}x} + 2x^2v^2 - 2x^2 = 0.$$

Since $x \neq 0$, we can divide by x^2, which gives

$$xv\frac{\mathrm{d}v}{\mathrm{d}x} + 2v^2 - 2 = 0,$$

as required.

Now we separate the variables. The equation says $xv\dfrac{\mathrm{d}v}{\mathrm{d}x} = 2(1 - v^2)$. Provided $v^2 \neq 1$,

$$\int \frac{v}{1 - v^2}\,\mathrm{d}v = \int \frac{2}{x}\,\mathrm{d}x \quad \Longrightarrow \quad -\tfrac{1}{2}\ln|1 - v^2| = 2\ln|x| + c.$$

Multiplying by -2 and taking exponentials gives $|1 - v^2| = Ax^{-4}$ for some constant $A > 0$, so

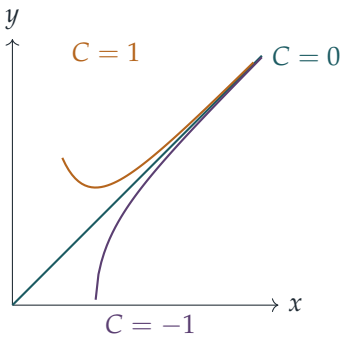
$$x^4(v^2 - 1) = C,$$

where $C = \pm A$ is a nonzero constant. The cases we set aside, $v = \pm 1$ (that is, $y = \pm x$), are also solutions: then $\mathrm{d}v/\mathrm{d}x = 0$ and $2v^2 - 2 = 0$. They are exactly the case $C = 0$, so C can be any constant.

Finally, put $v = y/x$:

$$x^4\left(\frac{y^2}{x^2} - 1\right) = x^2(y^2 - x^2) = C,$$

which is the required form.



The sketch shows a few solution curves in the first quadrant. They approach the line $y = x$ as $x \to \infty$, as $y^2 = x^2 + C/x^2$ suggests.

Part (ii)

Every term in $6x + 5y$ has degree 1, and so does $y\,\mathrm{d}y/\mathrm{d}x$. This suggests the same substitution, $y = xv$:

$$xv\left(v + x\frac{\mathrm{d}v}{\mathrm{d}x}\right) + 6x + 5xv = 0.$$

Dividing by $x \neq 0$ and collecting terms,

$$xv\frac{\mathrm{d}v}{\mathrm{d}x} = -(v^2 + 5v + 6) = -(v+2)(v+3).$$

Ask yourself How do we integrate $\dfrac{v}{(v+2)(v+3)}$?

Use partial fractions. Write $v = P(v+3) + Q(v+2)$. Putting $v = -2$ gives $P = -2$, and putting $v = -3$ gives $Q = 3$. So

$$\frac{v}{(v+2)(v+3)} = \frac{3}{v+3} - \frac{2}{v+2}.$$

Provided $v \neq -2, -3$, separating the variables gives

$$\int\left(\frac{3}{v+3} - \frac{2}{v+2}\right)\mathrm{d}v = -\int\frac{\mathrm{d}x}{x} \quad\Longrightarrow\quad 3\ln|v+3| - 2\ln|v+2| = -\ln|x| + c.$$

So $|x|\,|v+3|^3/(v+2)^2$ is constant. Removing the modulus signs (the sign of $x(v+3)^3$ can't change along a solution curve on which $v \neq -3$), we get

$$\frac{x(v+3)^3}{(v+2)^2} = K.$$

Now substitute $v = y/x$. The numerator is $x(y+3x)^3/x^3$ and the denominator is $(y+2x)^2/x^2$, so the powers of x cancel completely:

$$\boxed{(y+3x)^3 = K(y+2x)^2},$$

where K is a constant.

The lines we excluded are also solutions. If $y = -3x$, then $y\,\mathrm{d}y/\mathrm{d}x = 9x$ and $6x + 5y = -9x$, so the equation holds. This line is the case $K = 0$ of the boxed form. If $y = -2x$, then $y\,\mathrm{d}y/\mathrm{d}x = 4x$ and $6x + 5y = -4x$, so this line is a solution too. It isn't covered by any finite K; it corresponds to $K \to \infty$. Writing the solution as $(y+2x)^2 = k(y+3x)^3$ would cover it instead, with $k = 0$.

Checks

- Part (i): differentiating $x^2y^2 - x^4 = C$ implicitly gives $2xy^2 + 2x^2y\,y' - 4x^3 = 0$. Dividing by $2x$ gives back the original equation.
- Part (ii): we stepped the equation numerically (with a Runge–Kutta method) from many random starting points. Along each solution, the quantity $(y+3x)^3/(y+2x)^2$ stayed constant to about 10^{-14}. For example, the curve through $(1,1)$ has $K = 64/9$.
- The straight lines through the origin that solve each equation, $y = \pm x$ and $y = -2x$, $-3x$, come from the roots of $2v^2 - 2 = 0$ and $v^2 + 5v + 6 = 0$. That is a good sign that the v-equations are right.

Common slips

- Forgetting the product rule: $\mathrm{d}y/\mathrm{d}x$ is $v + x\,\mathrm{d}v/\mathrm{d}x$, not $x\,\mathrm{d}v/\mathrm{d}x$.
- Getting the partial fractions wrong (say, as $2/(v+2) - 3/(v+3)$). Test the decomposition at a value such as $v = 0$.
- Dividing by $1 - v^2$ or $(v+2)(v+3)$ without remarking that this loses some straight-line solutions.

Beyond STEP

An equation $\mathrm{d}y/\mathrm{d}x = F(y/x)$ is called homogeneous. Its solution curves are mapped to one another by the scalings $(x, y) \mapsto (\lambda x, \lambda y)$, which is why $v = y/x$ works. This is the simplest case of a deep idea, due to Sophus Lie: a symmetry of a differential

equation can be used to reduce its order. The straight-line solutions $y = mx$ are the curves that the scaling leaves fixed. Boyce and DiPrima, *Elementary Differential Equations*, covers homogeneous and other first-order equations.

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